

Marco Ciccone

PERSONAL INFORMATION

Surname Name: Ciccone Marco
Nationality: Italian
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RESEARCH INTERESTS

Deep Learning, Meta Learning, Transfer Learning, Continual Learning,
Game Theory, Multi-Agent Learning, Computer Vision

EDUCATION

Politecnico di Torino, Italy

Postdoctoral Researcher
Advisor: Prof. Barbara Caputo

June 2021 - now

Politecnico di Milano, Italy

PhD in Information Technology

April 2021

- Thesis: “*On Iterative and Conditional Computation for Visual Representation Learning*”
- Advisor: Prof. Matteo Matteucci
- Co-Advisor: Phd Jonathan Masci
- Grade: *cum Laude*

Romanian Institute of Science and Technology, Cluj-Napoca, Romania

Visiting PhD student, advised by PhD Luigi Malago’

May-July 2018

“*Information Geometry of Adversarial Robustness of Neural Networks*”

Politecnico di Milano, Italy

MSc in Computer Science and Engineering

April 2016

- Thesis: “*Pixelwise Semantic Segmentation of urban scenes using Recurrent Deep Neural Networks*”
- Advisors: Prof. Matteo Matteucci, Phd Francesco Visin
- Language: English
- Grade: *110/110 cum Laude*
- GPA: *28.85/30*

Università degli Studi di Firenze, Italy

B.Sc. in Computer Engineering

July 2013

- Thesis: “*Blind Source Separation methods for multichannel audio signals based on Independent Component Analysis*”
- Advisor: Prof. Fabrizio Argenti
- Language: Italian
- Grade: 97/110

PUBLICATIONS

- [1] F. Cacciamani, A. Celli, **M. Ciccone**, and N. Gatti, “Multi-agent coordination in adversarial environments through signal mediated strategies,” in *Proceedings of the 20th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2021
- [2] M. Cannici, **M. Ciccone**, A. Romanoni, and M. Matteucci, “A differentiable recurrent surface for asynchronous event-based data,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 1–17, Springer, 2020
- [3] M. Cannici, **M. Ciccone**, A. Romanoni, and M. Matteucci, “Asynchronous convolutional networks for object detection in neuromorphic cameras,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2019
Best paper award, Second International Workshop on Event-based Vision and Smart Cameras Workshop at CVPR2019
- [4] M. Cannici, **M. Ciccone**, A. Romanoni, and M. Matteucci, “Attention mechanisms for object recognition with event-based cameras,” *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2019
- [5] **M. Ciccone**, M. Gallieri, J. Masci, C. Osendorfer, and F. Gomez, “Nais-net: Stable deep networks from non-autonomous differential equations,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018
- [6] F. Lattari*, **M. Ciccone***, M. Matteucci, J. Masci, and F. Visin, “Reconvnet: Video object segmentation with spatio-temporal features modulation,” *The 2018 DAVIS Challenge on Video Object Segmentation - CVPR Workshops*, 2018
- [7] A. Romanoni, **M. Ciccone**, F. Visin, and M. Matteucci, “Multi-view stereo with single-view semantic mesh refinement,” in *Proceedings of the IEEE International Conference on Computer Vision Workshops (ICCVW)*, pp. 706–715, 2017
- [8] F. Visin, **M. Ciccone**, A. Romero, K. Kastner, C. Kyunghyun, Y. Bengio, M. Matteucci, and A. Courville, “ReSeg : A Recurrent Neural Network-based Model for Semantic Segmentation,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2016
Best paper award, DeepVision Workshop at CVPR2016

TEACHING

Mediterranean Machine Learning School (M2L) 2021

Tutorial on Generative Models (VAEs)

January 2021

Deep Learning 2017/2018, Prof. Matteo Matteucci

PhD and MSc Course at Politecnico di Milano

Teaching Assistant

February 2018

- Tensorflow and Pytorch Tutorial
- Advanced Convolutional Neural Networks

Cognitive Robotics 2016/2017, Prof. Matteo Matteucci

MSc Course at Politecnico di Milano

Teaching Assistant

June 2017

- Deep Learning Introduction
- From Linear Models to Deep Neural Networks
- Convolutional and Recurrent Neural Networks

REVIEWER

Neural Information Processing Systems (NeurIPS2021)
International Conference on Learning Representations (ICLR2021)
AAAI Conference on Artificial Intelligence (AAAI2020, AAAI2021)
International Journal of Computer Vision (IJCV)
International Conference on Robotics and Automation (ICRA2017)
International Conference on Intelligent Robots and Systems (IROS2017)
Geometric meets Deep Learning Workshop (ICCV2017)
Neural Computing and Applications Journal (NCAA)
Geometric Deep Learning Workshop (ECCV2018)

CONFERENCES AND
COURSES

Transylvanian Machine Learning Summer School 2018, Cluj, Romania

Acceptance Rate: 8%

16-22 July 2018

Machine Learning Summer School 2017, Tübingen, Germany

Acceptance Rate: 13%

19-30 June 2017

International Computer Vision Summer School 2017, Sicily, Italy

Acceptance Rate: 35%

9-15 July 2017

**International Conference on Acoustics, Speech, and Signal Processing
ICASSP 2014**, Florence, Italy

Staff volunteer

4-9 May 2014

Attended tutorials :

- Informed Audio Source Separation: Trends, Approaches and Algorithms
- Deep learning for natural language processing and related applications

PROFESSIONAL
EXPERIENCE

NVIDIA, Santa Clara, California

Research Intern

August 2018 - November 2018

Project on adversarial robustness of Neural Networks.

NNAISENSE, Lugano, Switzerland

Research Intern

July 2017 - January 2018

I spent 7 months as research intern at NNAISENSE tutored by Jonathan Masci and Marco Gallieri. In this time I worked on stability and robustness of Residual Networks from a control theory perspective.

Horus Technology, Milan, Italy

Machine Learning Engineer

June 2016 - June 2017

Horus Technology is an Italian start-up (recently rebranded as Eyra) that operates in the field of Artificial Intelligence. Its first product, Horus, is a wearable device that aims to assist blind and visually impaired people thanks to state-of-the-art

computer vision technology. During my time here, I worked on deep learning models for Object Recognition and Scene Analysis to be deployed on a low power embedded system based on NVIDIA TK1.

Projest S.p.A., Prato, Italy

IT Consultant - Software Engineer

Nov 2009 - Oct 2012

I carried out several consulting projects as IT consultant for Guccio Gucci S.p.A., Gucci Logistica S.p.A. and other clients.

LANGUAGES AND
CERTIFICATES

English: Highly proficient in spoken and written English

Italian: Native speaker

French: Basic communication skills